Development of a Mobile Contraception Decision Aid for Transgender and Gender-Nonconforming Individuals Assigned Female at Birth

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1 Introduction

- Transgender and gender-nonconforming (TGNC) assigned female at birth (AFAB) individuals face unique challenges in selecting contraceptives.
- We report an ongoing study to develop a mobile contraception decision aid to support the TGNC AFAB community for sexual health education and contraception decision assistance.
- This work addresses limitations of a desktop tool developed from a pilot study to improve accessibility, gender inclusivity, system functions, and usability.

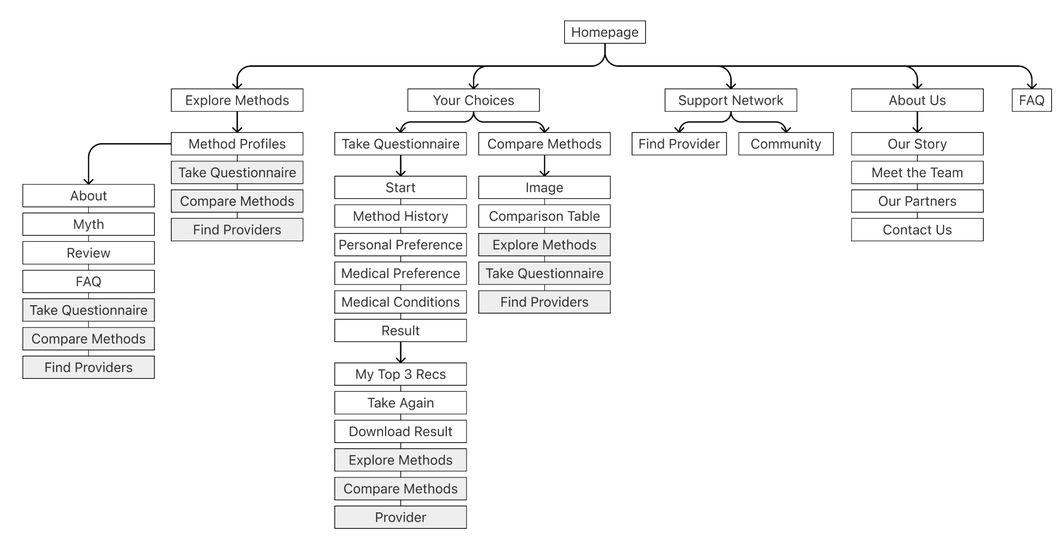
2 Methods

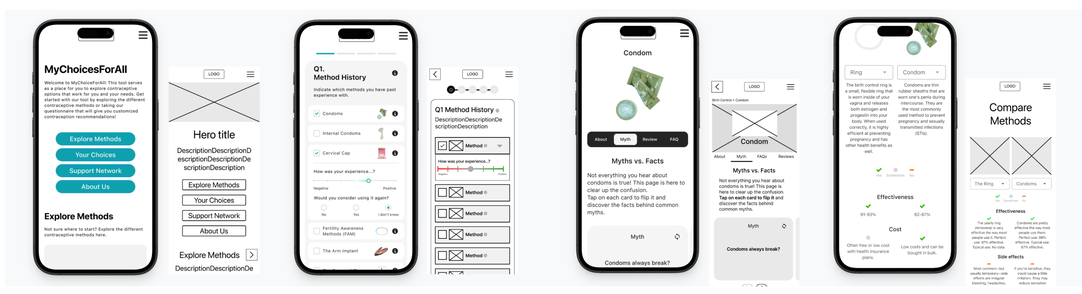
- We leveraged user-centered design principles, with techniques such as mental models and ethnographic data, for system design.
- We used Figma to create wireframes with multiple iterations to yield a working mobile prototype.
- The implementation utilized Node.js for the backend and Next.js, TypeScript, and Tailwind CSS for a responsive, well-structured frontend.
- We are now testing the deployment of the system on AWS for reliability and scalability.

3 Results

- We reorganized the tool with three main features:
  (1) Exploring specific contraceptives, with description of each method, common misunderstandings, and FAQs;
  (2) Personalized recommendations of contraceptives based on medical conditions and individual preferences, which can then be used further for discussion with healthcare providers; and
  (3) Comparing contraceptive methods with a list of decision factors.

- We redefined the brand style by using simple, gender-neutral design languages and incorporated more visual elements to enhance accessibility.
- We provided outlines for new features such as review/experience of contraceptives and TGNC AFAB support network.



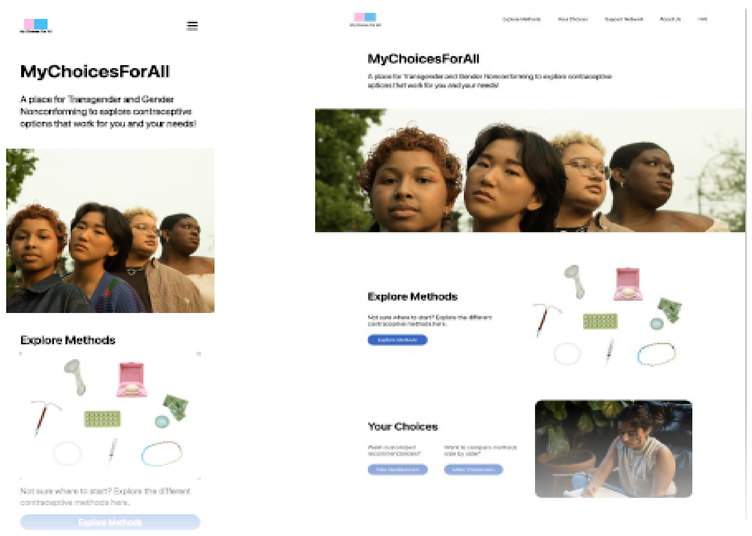


4 Discussion

- In development of the mobile tool, we focused primarily on improving system design by adding new features and enhancing the existing functions.
- We are now conducting studies to assess system function, usability, and trust, developing an integrated system through responsive design, and enhancing the scoring algorithm for customized recommendations to better support the TGNC AFAB community.

5 Current Status

We are iterating the site based on findings from our latest research, focusing on refining the visual design, improving navigation, and expanding health information.



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